

CHAPTER FIVE

The Application Process: Where Theory Meets Reality

In this chapter we move from intellectual underpinnings to the practical application of training theory. This will take you one step closer to and better prepare you for making your own plan.

One thing should be abundantly clear after reading the earlier sections of this book: Top endurance athletes have maximized their aerobic capacity. This means that their AeT is pushed right up close to their LT. Remember the 10 percent test? The LT, in turn, is pushed right up near the maximum heart rate. The following two graphs are made from the lactate tests of one athlete. Figure 5.1 comes after a five-month base training season that included a high percentage of Zone 3 and Zone 4 training with very little time spent in Zone 1 or Zone 2. Figure 5.2 is after a five-month base training season using the method described in this book.

For every event covered in this book—from the speediest Vertical Kilometer, to a 100-mile run, to a multiday ski tour—the biggest limitation to performance will come from your aerobic metabolism. Hence we will focus most of our attention there. All other training will be acting in a supplemental and supportive role to enhance that aerobic development.

So how do you develop this essential quality for yourself? Using the models and principles previously introduced, we're now going to explain the various methods available to maximize your aerobic metabolic output.

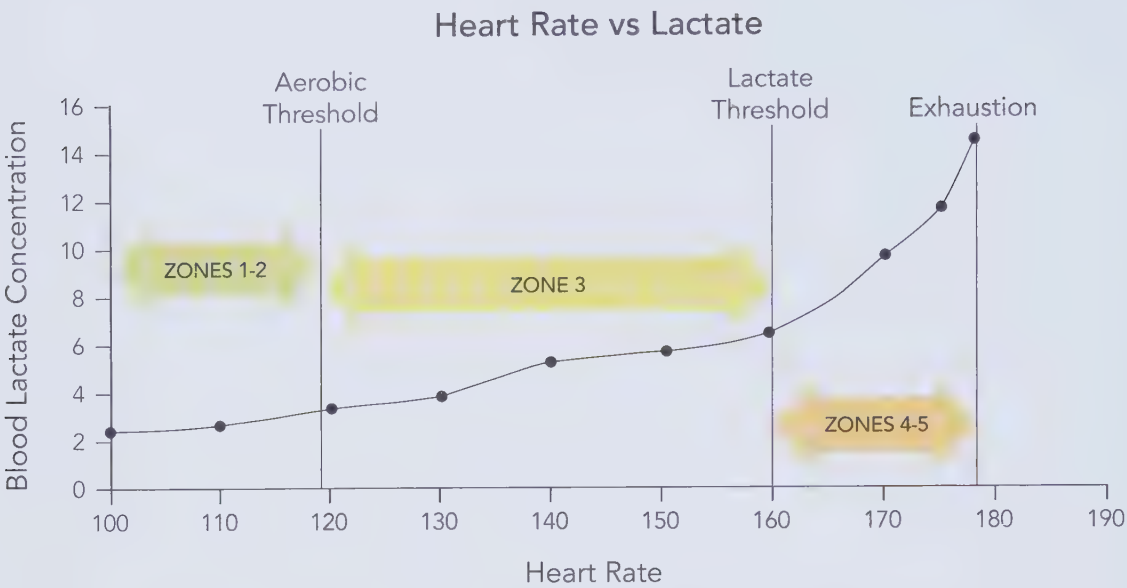


Fig. 5.1 Heart Rate Versus Lactate in Low Aerobic Capacity Athlete

Here we see the results of a blood lactate test for an athlete with low aerobic capacity. It shows the AeT (the top of Zone 2) occurs at a heart rate of around 120 bpm, which corresponds to a running pace of about 10 min/mile (6:12 min/km). The top of Zone 3, or the LT, occurs at a heart rate of 160 bpm, corresponding to a pace of 8:20 min/mile (5:10 min/km). With this forty-beat spread between thresholds, this athlete has Aerobic Deficiency Syndrome and needs to do considerable aerobic base work.

Testing

Establishing your own body’s metabolic response to exercise is important if you want the best plan. And since we are talking application now, testing is the logical place to start.

Aerobic Threshold Testing

Determining your AeT establishes the most critical metabolic reference point you can know and serves as the foundation from which to build your training plan. We recommend that you know this intensity. Several methods exist and we’ll discuss them in order of ease of implementation.

1) MAF method. The Maximum Aerobic Function is a simple method developed by Dr. Phil Maffetone in the early 1980s that has gained considerable acceptance in the ultrarunning world. It is a simple formula: subtracting your age from 180 if you’ve been training consistently for two years without significant setbacks. Maffetone suggests the following modifiers based on your training history.

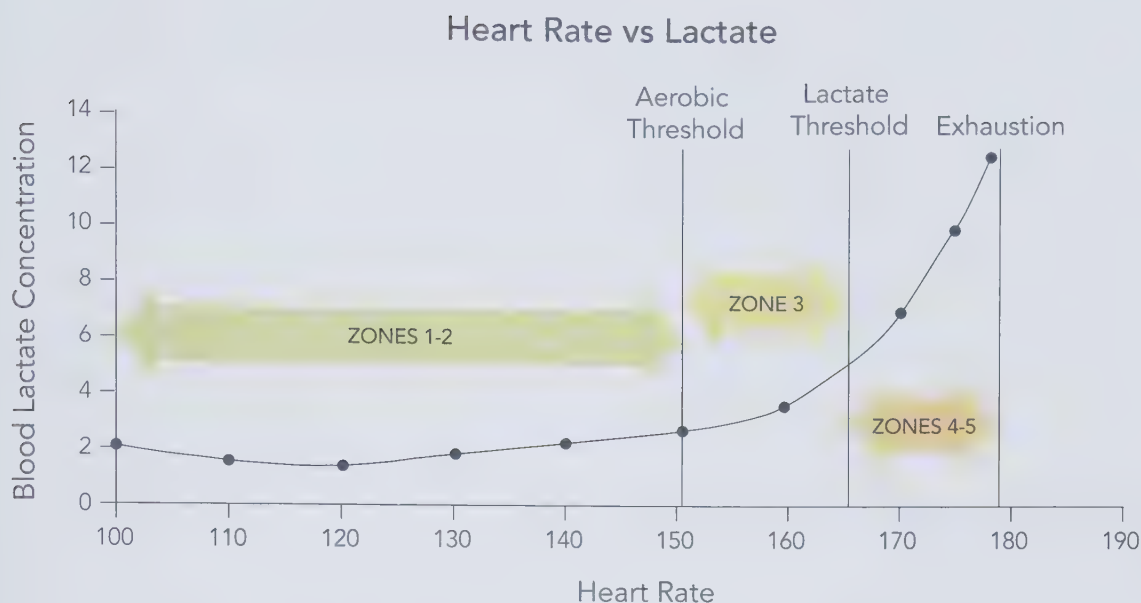


Fig. 5.2 Heart Rate Versus Lactate in a Well-Trained Athlete

Above we see the results of a blood lactate test for the same athlete after a season of proper training. The graph shows the large increase in aerobic capacity. Notice that the AeT has moved up to 150 bpm and corresponds to a pace of 8:15 min/mile (5:09 min/km) and the LT has bumped to 165 bpm and a pace of 7:30 min/mile (4:40 min/km). In this condition this same athlete can now sustain a pace for several hours that he could previously struggle to manage for even one hour. Remember the vacuum analogy? This is what a powerful vacuum looks like.

- If you're recovering from an injury or long layoff or get sick more than two to three times per year, subtract an additional five.
- If you've been training for more than two years and showing regular progress, then add five.

In our experience, this formula does tend to err on the side of being conservative; it tends to set your AeT heart rate limit a bit low, which makes it a safe method especially for newcomers and the less fit. Its biggest problem is that it can't account for increases in your AeT heart rate. Our experience shows that people who are new to this type of aerobic base training program will see a progressive increase in their AeT pace and the corresponding heart rate. With the MAF formula, you are stuck with a heart rate derived from a chronological model.

2) Ventilatory marker. In hundreds of field tests in moderate-to-well-trained endurance athletes, we have seen a good correlation between the heart rate that can be maintained while breathing only through the nose or while carrying on a conversation with the blood lactate level of 2mmol/L (the commonly accepted



Day 30 ... or maybe 32 ... of Der Lange Weg (The Long Trail). David Wallmann, in the distance, and Philipp Reiter ski into the first light of day, France. Photo: Mark Smiley

measure of AeT). The caveat here is that this works only for those with good aerobic training backgrounds. For those who have no history of aerobic endurance training, or those who have a training background heavily biased toward high-intensity, short-duration work, or people with no history of regular exercise, these ventilation markers fail to correlate with blood lactates. If that describes you, then use the MAF method or read on.

3) Lactate step test. This test can be administered using a treadmill set to a 5 percent gradient or outside on a moderate hill. Start with a fifteen-minute warm-up to enable the aerobic system to come fully online. The test itself consists of a series of three-minute stages at increasing speeds that are just enough to elicit a bump in heart rate of about ten beats per minute. It is important to get the heart rate to stabilize as soon as possible at each new stage and hold it as steady as possible. At the end of each stage, a small blood sample is taken and measured using a portable lactate analyzer to determine the lactate concentration present in the blood. The test needs to be started at what might appear to be an unnecessarily low heart rate—probably walking—and very gradually increasing it ten beats per stage so that you slowly sneak up on the AeT. The heart rate of the AeT is reached when the lactate level reaches or exceeds 2mMol/L, or the first stage where the lactate number increases by 1mMol/L. If using a hill, simply walk back down between stages.

This test is also easy to administer to yourself or with a friend; for less than the cost of lab tests, you and a friend can buy your own lactate meter and do these tests whenever you want. Here is a great little meter we use: <http://www.lactate.com/lactateplus.html>.

4) Gas exchange test. This test is done in a lab on expensive equipment run by technicians. These tests can range from \$150–\$350 but are the gold standard for determining your metabolic response to exercise. They can determine your AeT, LT in terms of heart rate and pace (albeit on a treadmill), and, if taken to failure, your VO_2 max. Going to failure is unnecessary for the purpose of finding the two thresholds. Many labs will supply an easy-to-read graph of fat versus carb usage at various intensities. The lab tech should be able to explain the ramifications of these test results. The most valuable and actionable information from the test will be to see at what heart rate you reach your AeT and where it lies in comparison to your LT. If the AeT is more than 10 percent below the LT in terms of heart rate, then you can still make considerable gains in raising the AeT.

Caveat: Many labs and fitness facilities hawking physiological testing are unfamiliar with the AeT test or even the concept of AeT and will want to administer a VO_2 max test and along the way tell you where your LT occurs. This relates back to the sidebar about VO_2 max earlier in the Physiology Section. The problem with using this type of test for determining your AeT is that the VO_2 max test protocols often use short-intensity stages. This will have you zooming quickly through your aerobic zone because they are in a rush to get you to redline and measure VO_2 max—their goal—before you tire. They move through the aerobic zone so fast that your metabolism never stabilizes. To determine the aerobic response to exercise intensity, you need at least three-minute exercise stages.

Lactate (or Anaerobic) Threshold Testing

Unlike the AeT test, this one is fairly straightforward. It involves a maximum sustained uphill effort of thirty to sixty minutes' duration. A treadmill can suffice as a second choice. The less well trained should use thirty minutes of sustained effort, otherwise muscle fatigue will limit your ability to sustain a hard effort.

Start with a good warm-up of at least fifteen to twenty minutes. Do not test this when you have tired legs. Reduce training load for a few days beforehand to get the most indicative results. Finish the warm-up with two to three minutes of Zone 3 running or skiing. Start your recording HR monitor as you begin your test. Try to quickly settle into a pace that is maximum but sustainable for the full test. If you are attentive to your effort and heart rate, you will notice that your heart rate will settle in a very narrow range that feels comfortably hard. If you speed up even by two to three beats per minute you will be forced to slow again



Luke Nelson embracing discomfort while diving into the difficult later stages of threshold testing.
Photo: Greg Snyder

very soon. Try to find this sweet spot and hold that intensity for the full test. At the end, your average heart rate will indicate your LT intensity (top of Zone 3). Well-conditioned athletes will be able to sustain this output for up to an hour. But less-fit individuals will fatigue, which is why we recommend the thirty-minute test length for them.

Aerobic Base Development: Volume Is the Key

You will never maximize your endurance without first maximizing your basic aerobic capacity (AeT).

While genetic gifts play the dominant role in preselecting who is going to be good at power and speed sports, perseverance and the ability to suffer are the hallmarks of the successful endurance athlete. No wonder endurance sports appeal to the type A personalities for whom more equates to better. A question we often get is, “Just how much is enough?” That is an impossible one to answer. What might not even count as much of a warm-up for Kílian, something he would do before breakfast on a recovery day, might overload others. It all depends on your capacity for this type of work.



Putting Some Numbers to It

Figuring out your optimal training load is a highly individualized determination. We give some guidelines in the Programming section of this book, but the best advice is to start small and work up rather than to bite off more than you can handle. When launching into a new season of training with a new plan, you'll be excited and hopefully well rested. Just as in a race, it is all too easy to go out too hard in the short term only to find that you overextended yourself in the first few weeks. This can be demoralizing and require you to rethink your plan. It is much easier to adjust the training load upward after you determine that it is too light than it is to crumple, dejected or injured, from doing too much too soon. It is hard to progress if you have to move backward.

How can you tell you are doing the right volume of aerobic training? Here's a simple test:

When it comes to basic aerobic training (Zones 1 and 2): If you can't do the same workout you did yesterday again today, and tomorrow and the day after for days on end, then you are doing too much. You are doing too much volume or too much intensity for your current regimen to be considered an aerobic base-building program. Caveat: If the top of your Zone 2 is within 10 percent of the top of your Zone 3, you will need to be more sparing with the Zone 2 training. For you, the first sentence in this paragraph would include only Zone 1.

World-class endurance athletes spend a prodigious amount of time training for their events. At the highest levels it is truly a full-time job, allowing for only enough time and energy to recover between workouts. Most of us won't have that luxury, but to put things in perspective let's look at the training of a nationally competitive mountain runner and skimo racer with a full-time job and family. These are weekly averages for a twelve-month period including a three-week break from training:

- Average hours training time per week: 12.5 hours. With three weeks over 40 hours, and 11 weeks over 23 hours. Annual total = 650 hours.
- Average elevation gain and loss per week: 9,843 feet (3,028 meters). With six weeks over 20,000 feet (6,154 meters), and 17 weeks over 15,000 feet (4,600 meters). Annual total = 512,000 feet (157,538 meters).
- Average running distance per week: 55 miles (90 kilometers). With seven weeks over 100 miles (160 kilometers), and 25 weeks over 75 miles (120 kilometers). Annual total = 2,860 miles (4,576 kilometers).

At the top end, Kilian trains in the range of 1,300 hours per year and over 1.5 million vertical feet. Keep this in mind when you read of his amazing feats of



Chad Sayers ascending the southwest ridge of Mount Shkhara, Caucasus Mountains, Republic of Georgia. Photo: Jason Thompson

speed and endurance. You are seeing what Dr. K. Anders Ericsson, the psychologist and world expert on excellence, calls the iceberg illusion: the end product of a preparation process measured in years. Whether or not Kilian has some special genetic gifts is immaterial. What matters is that he has, since childhood, created a structure in his life where he could devote himself fully to training.

The point we want to make with these figures is not to intimidate you or to suggest that you must train at this level to succeed. *Base your expectations on the reality that your results will be directly proportional to the time you spend in preparation.* A trio of forty-five-minute stair machine workouts a week will not allow you to reach your ultimate potential. But this may be a perfectly reasonable starting place for you, given your personal constraints. The important thing is to start from the right point, and then progress, adapt, and in the process, learn some valuable lessons.

So, why not just do what the pros do? Because you will not have the work capacity of a pro and will likely end up overtrained and/or injured if you merely copy a world class-level training program. The human body has an amazing capability of adapting to imposed loads if it is allowed sufficient time for the adaptation to occur. There is no secret or magic. It is just a work ethic and building a lifestyle around that work ethic.

Getting Started

When you first embark on a program of aerobic development that has as much as 90 percent of endurance training done at or below your personal AeT, it often provokes some questions and comments. While an elite runner is not running slowly when he is in his low-intensity zones, the sad reality is that for many athletes, their Zones 1–2 training will be an excruciatingly slow pace that seems like it cannot possibly have any beneficial training effect at all. If you find yourself in this situation, welcome to the club: the Aerobic Deficiency Syndrome (ADS) Club (see ADS discussion on page 46). The reason your AeT pace is so slow is that your aerobic capacity is low. In other words, your aerobic metabolism can crank out only enough ATP to support what may be a disappointingly slow pace. The only way to increase your speed at an aerobic level of intensity is to increase your aerobic capacity, which means you must train at and below your AeT.

If you have ADS, it is going to be possible for you to, initially, do almost all of your aerobic training at or very close to your AeT or top of Zone 2. The reason for this is that the pace will be slow enough to not tax your muscles. The problem that most people will have is going slow enough. They have become used to training at a pace usually well above their AeT. The common mistake of trying to constantly push the endurance limit up by training right at the LT is almost an epidemic in the endurance training world. Getting used to slowing down to an aerobic pace is going to take a lot of discipline for most people. But there is no shortcut and without this, the whole aerobic training house will be built on a very shaky foundation.

Remember from our previous examples that *low intensity* does not mean *slow* in well-conditioned athletes. Those who have spent the time to maximally develop their aerobic capacity can maintain prolonged high speeds at their AeT.

So, the first step is to get that aerobic base built as well as you can manage, given your time constraints, both in terms of your annual plan and your daily schedule. Just remember: when in doubt, go long and easy.

There is no prescription suitable for everyone. Some athletes with a very high aerobic capacity will be able to run 100 miles a week at a 7-minute-per-mile pace (4:20 per km). Others will get out of breath on the slightest uphill grade, be forced to walk to maintain an aerobic pace, and will struggle with 20 miles a week.

By maintaining a steady diet of this training for months on end, the components of your aerobic system will become much more powerful. You can recheck your aerobic development every two weeks or so by doing your own AeT time trial. Since this will be relatively easy, you can just toss it into a regular workout. Simply note your time at the same heart rate over one of your regular courses so you can compare it in a few weeks. If training is being done correctly, you will see encouraging drops.

Just remember that when you close the AeT-to-LT gap to within 10 percent, you'll need to reduce the amount of Zone 2 training and do more Recovery and Zone 1 workouts. Do not feel that if you are not running right at your aerobic limit, you are not getting the aerobic benefits. Now you are ready to introduce some Zone 3 and Zone 4 training.

Endurance Training Methods

Finding the balance between long duration, low-to-moderate-intensity training and short-duration, higher-intensity training is one of the things that makes coaching and training more art than science. While we've explained some useful general guidelines, the balance of this mix depends upon the event being trained for (specificity) and the individual being trained (individuality). Shorter events need more high-intensity training. More seasoned athletes need more high-intensity training.

Two distinct methods exist for improving endurance: continuous and intermittent (interval) training. We'll delve into each in this section.

Continuous Training for Improving Basic Aerobic Capacity

By continuous, we mean a steady-state exercise lasting for a minimum of thirty minutes and up to several hours. This is how you'll be accumulating all those hours in Zone 1 and Zone 2 that need to make up about 80 to 90 percent of your total annual training volume. Don't let the term *basic* in the title above mislead you into thinking this stuff is basic and you can skip right past it. That is not the kind of *basic* we are referring to. We are using the term here to mean a base of support for any other training to come on top of. Without this base, the rest of the training will never be as effective. Even a thirty-minute morning jog can be beneficial for maintenance. Keep in mind the role *continuity* plays: These low-intensity, continuous workouts need to be done often and regularly to have their intended effect.

WARM-UP

This seemingly simple task is so misunderstood and misapplied that we feel the need to lay out some prescriptive terms and guidelines to be followed in the warm-up process. The warm-up serves the purpose of preparing the muscles to do the workout in their optimal state. As a general rule, the higher the intensity





ATHLETE STORIES

Tracing My Own Lines

Luke Nelson

Jared and I were about thirty-five hours into our Nolan's 14 Fastest Known Time (FKT) attempt, and I was intent on quitting. I couldn't imagine all the ground we still had left to cover after what we'd already done.

Earlier in the day, when we were standing on the fifth or sixth summit, I asked Jared to point out the last mountain we would summit. "You see that mountain on the horizon?" he said. "It's past that." My mind exploded—I could barely see the peak clear out on the horizon, and the one we'd end on was even farther away.

There are fourteen fourteeners in the Sawatch Range in Colorado, and for Nolan's 14 you start at one end and pick your own route through the range, summiting each of the peaks. It's about 100 miles with 46,000 feet of ascent. In September 2017, Jared Campbell and I set out to break the existing record for this do-it-yourself traverse.

I've always been drawn to moving efficiently in the mountains. I didn't come to running from a running background. I started mountain running about ten years ago, and before that I traveled through the mountains by other means: I kayaked, climbed, back-country skied, and snowboarded. Movement in the mountains has always been the most intriguing part of endurance to me.

For several years my main focus was trail racing, but there were always these secondary mountain projects calling to me. They remained secondary because I had to taper for a race or recover from a race. I'd do them between races or would use them as long training runs. Then I made one a primary focus: in August 2014, Jared and I set the FKT for the 12,000-foot peaks in Idaho. We climbed all nine of them in a single push. The previous FKT was thirty-eight hours, and we cut that down to twenty-eight hours.

I began to feel more of a desire to channel my race-type efforts into FKTs and moving on my own in the

mountains. This time of soul-searching came down to a pivotal discussion with my coach, Scott Johnston. "Look, you have to make a decision between doing these amazing adventures and racing, because you can't do both," he said. "It's just too much to do well."

Pulling back from racing is a hard thing to do as a professional mountain runner, because that's where the community comes together, and that's how your fellow racers and your sponsors gauge your ability. To march to your own drummer is tough. At first I was really nervous, because I didn't know what my sponsors' reaction would be, but the companies I work with have been extremely supportive.

I made projects like the Idaho linkup the theme of my summer in 2017. Jared and I did a multiday crossing of what was the Bears Ears National Monument, about 140 self-supported miles in three days, and then we followed that up with an FKT of all the 13,000-foot peaks in Utah in a single push. There are nineteen of them, and we did all of those in thirty-three hours. Jared and I dream big—sometimes too big.

I love that you can have a loose date range for an independent objective. You can wait for things to align perfectly—whether that's training, weather, or conditions—and get after it when everything is ideal. With Nolan's, the date range was August to mid-September. As soon as it looked like the monsoon season was coming to an end and the weather window was opening up, we nailed down a date, did our final prep, and went for it.

Out there on Nolan's, when I decided I was going to quit more than a full day in, Jared played some brilliant mental chess to get my mind going again in the right direction. We spent hours moving forward, Jared listening to me insist I would drop when we reached my dad, who was crewing.

At last we made it to my dad. Jared took off his shoes. Then he started eating some food. Five minutes ticked by.

"Hey, man, let's keep going," I said. I understood that even though I wasn't mentally prepared for the task, physically I was. It doesn't matter whether or not you can imagine how far you still have to go; all you have to do is keep putting one foot in front of the other.

We rallied and got it done. We didn't break the record, but we have the second-fastest time in 53:29.

While trail racing is super-fun, and races are great places to gather with the community, I see far more potential to do very interesting things moving independently in the mountains as opposed to just going to a race. At a race you are tied to a course. You are limited by someone else's imagination—by what they think is difficult. One of the biggest catalysts to my transition away from racing was that I was looking to draw my own lines in the mountains, to follow a little different path than most.

Nolan's is the most difficult thing I've ever done. The sleep deprivation, the altitude, the off-trail travel—all of those things combined in 100 miles was pretty amazing. It goes to show that most of us can do more than we think we're capable of. That's a recipe for an incredible adventure: when it exceeds what's considered possible. ■

Luke Nelson has spent his life wandering the mountains. After climbing, kayaking, and snowboarding throughout his youth, he discovered running just over a decade ago as a way to move through the high country. He has dozens of podium finishes in mountain running races and ski mountaineering races, including winning a US Ski Mountaineering National Championship. Overall, what he enjoys most is moving quickly and efficiently through the mountains.



Luke Nelson is all smiles after completing a two-day traverse of the Boulder-White Clouds Range. Ketchum, Idaho. Photo: Steven Gnam

of the workout or race, the longer the warm-up. For workouts in Zones 1 and 2, a perfectly good warm-up can be accomplished by easing into the pace of the workout for ten minutes. However, if you are doing high-intensity training, we suggest following a warm-up routine outlined below. On days when you are tired, allow a longer time for the warm-up to assess how you are feeling and if any adjustments need to be made to the day's planned workout.

Stage 1: Your body will have likely been in a fairly sedentary state before beginning the warm-up. As such, the parasympathetic portion of the autonomic nervous system will have been dominating. This is the portion of your nervous system that drives the resting and digesting functions. This part of the warm-up serves a purpose of getting the sympathetic nervous system activated. This dilates the capillaries in the skin and muscles and adjusts blood pressure accordingly so that the working muscles can get their full complement of blood and oxygen. Raising the body temperature is a key part of the warm-up. This is best accomplished with light aerobic exercise. Notice when you begin to break a sweat. That is an indication of the capillary dilation and elevated temperature. Expect this to take at least ten minutes. This period also serves the purpose of getting the aerobic system fully online.

Stage 2: Limbering callisthenic movements and dynamic stretching done at a gradually increasing pace and effort will begin to activate some of the stretch reflexes in the joints and prepare your muscles for higher loads. You should not feel any strain during this portion of your workout. This stage can last from five to ten minutes. If you are moving into an interval session, finish this phase of the warm-up with two to three minutes in Zone 3 using the training mode of the main workout (running or skiing).

COOL-DOWN

The purpose of the cool-down is to gradually return your body's systems to homeostasis, reduce the core temperature, metabolize some of the waste products, and relax psychologically from the focus of the workout. The higher the intensity of the workout (or race), the longer the cool-down should be. A long, low-intensity workout may not require any cool-down other than a couple of minutes' walking at the end. After a high-intensity workout or race, a typical cool-down should be twenty to thirty minutes of the Recovery Zone or Zone 1.

RECOVERY WORKOUTS

The importance of these very low-intensity workouts cannot be overstated. As your AeT improves, your Zone 2 workouts will impose higher neuromuscular loads and need to be spaced further apart. These lowest-intensity recovery workouts will begin to fill a bigger and bigger part of your training volume. You

Example Warm-Up Routines

- Easy aerobic running or skiing in Z1 until you sweat. Maybe 10 to 15 minutes.
- If a low-intensity workout is planned, move directly into that.
- If a high-intensity workout is planned, move to the next steps.
- For running, do the following exercises with gradually increasing intensity:
 - 10 air squats, no rest, then ...
 - 10 leg swings for each leg, no rest, then ...
 - 10 trunk rotations for each side, no rest, then ...
 - High-knee running in place for 20 seconds, no rest, then ...
 - Bouncing on your toes for 20 seconds, no rest, then ...
 - 2 to 3 minutes in Z3–Z4 uphill.
- For skimo, do the following:
 - 3 min uphill skiing in Z3, no rest, then ...
 - 2x10 sec uphill skiing in Z5 with 2-min recovery between.

will need more modulation of the training load. This light aerobic work has a dramatic restorative effect on a tired body.

The jury is out on the *why*, but almost all endurance coaches have seen the value of prescribing these mellow workouts. Author Johnston has coached very well-trained cross-country skiers who thrived on long recovery workouts. Several Western coaches of Kenyan runners have remarked about the Kenyans' dependence on very long, "stumblingly slow"–paced recovery runs as key elements of their training.

Most people will find thirty minutes to an hour a useful amount per workout. It may only be a twenty-minute brisk walk before bed. Whatever the underlying reasons, this very low-intensity training should not be dismissed as too easy to be doing anything worthwhile. At the very least it is preparing your body for the rigors of the upcoming workout by speeding the recovery process. If done properly, you should feel better at the end of this workout than when you started. And you should also feel much better the following day than you would have had you foregone this workout.

When you get in the practice of effective and consistent recovery workouts, you will notice that the next day's workout will almost always feel great. If it doesn't, then you either needed more recovery work or an easier pace. This is also the place to add in cross-training like biking and swimming, especially if your legs are feeling beat-up.

KEEP IT SIMPLE

Heart rate and ventilation will be your best guides here. It may mean slowing and even walking the steeper uphill to maintain the intensity at the desired level. Don't



Athletes in the Lagorai Cima d'Asta Ski Mountaineering Race. Photo: Federico Modica

worry that your heart rate drops on the downhills. Just let it drop. If you are running, don't thrash yourself on every downhill in a vain attempt to keep your heart rate in the zone. The downhills provide a valuable strengthening effect in mountain sports, so don't discount them just because your heart rate is low. We recommend counting your full time—both up and down the hills—for hilly workouts.

PICK-UPS

A very effective, and fun, workout is to include short pick-ups (speedy segments) into long Zone 1 and Zone 2 workouts. These pick-ups will range from eight to ten seconds in the earliest stages and up to fifteen seconds in the later stages. We could argue over why these work so well as a training stimulus, but suffice it to say that this strategy has nearly universal acceptance among endurance sports. Begin adding these after a few weeks of aerobic base training.

You may find that it suits you to do your pick-ups early in the workout when you are freshest and do them in a structured repetition style, or to toss them into the workout throughout as the terrain and feeling motivates you.

By doing these early in the workout, you'll be fresher and faster, developing more of a speed and strength training effect. By doing them late in a workout, you'll be asking your legs to push fast when fatigued and getting more of an endurance effect.

The volume of the pick-ups during an individual workout should initially be limited to no more than six to eight and the speed should be only a fun, fast pace—never a full sprint. Take at least two minutes of easy running between reps. We have found that a single workout per week of these is plenty. If done consistently in the base aerobic phase, these will do a good job of preparing you for the harder work to come.

HIGHER-INTENSITY CONTINUOUS WORKOUTS

These workouts serve a valuable purpose in training the mixed metabolic processes above your AeT and around your LT in Zone 3. As mentioned in the discussion of intensity zones, this training has a powerful effect on improving endurance and can be seductive. Don't fall into the trap of letting most of your distance runs morph into the following types.

Tempo

Tempo workouts involve continuous fun, fast efforts (Zone 3). They are usually from twenty to sixty minutes in length. Their intensity can span the full breadth of Zone 3 (between the AeT and LT). Remember that if you have done a good job of increasing your aerobic capacity, then your AeT will be within 10 percent of your LT. This will translate into a narrow Zone 3 heart rate range; often fewer than fifteen beats. The goal should be to hold as steady an effort/heart rate for the entire time. Resist the temptation to turn these workouts into hard efforts or personal time trials. That is not the purpose and in fact, going too hard will defeat the purpose, which (once again) is to hold a steady effort/heart rate. Use a ten to twenty-minute Zone 1 segment before and after the workout to warm up and cool down.

Also, well-conditioned athletes can benefit from adding a Zone 3 tempo segment (twenty to thirty minutes) in the middle of a long Zone 1/2 workout.

Progressive Distance

Progressive distance workouts have a similar effect as tempo workouts but with the added stress of doing the faster running at the end of a long workout. Start very slowly with a warm-up and gradually increase the pace or effort/heart rate during the workout. The main section of these workouts where you are operating just below and above your LT can last for ten to thirty minutes and will greatly enhance endurance due to your prefatigued state. These workouts can be done for as short as sixty minutes but become much more effective when extended beyond ninety minutes and even up to two hours.

Example progressive distance workout: Start with forty minutes in Zone 1. Then move to thirty minutes in Zone 2. By now you should have a sense of whether this is a good day to push harder or if it would be smart to back off and finish the workout in Zone 1, saving the hard work for a day when you feel better. If you do

feel good, add on twenty minutes in Zone 3. If still feeling strong, increase the effort for a final five minutes in Zone 4.

To get the full benefit, you will want to have one or two days of recovery workouts after these higher-intensity continuous workouts. One workout like this each ten to fourteen days after the early Base Period will be a sufficient stimulus even for the well trained.

SUMMARY

- Gauges of whether your aerobic intensity is appropriate include: Are you recovering overnight so that you can do the same workout again day after day? Can you carry on a conversation while doing these workouts? Can you breathe through your nose while doing these workouts? Is your blood lactate level staying below 2mMol/L while doing these workouts? Is your AeT pace increasing on a biweekly to monthly basis?
- Be patient. Do this unglamorous work well and you will be greatly rewarded later. Rush through this period, try to take shortcuts, or interject much long, high-intensity work during this time and you will be shortchanging your long-term chances before you even get into the racing season.
- Include pick-ups into one distance workout each week to begin to build some specific strength endurance before moving on to more demanding high-intensity aerobic interval training.
- After at least six to eight weeks of base training, you may find it useful to start adding in higher-intensity continuous workouts once every ten or so days.

High-Intensity Interval Training for Maximizing Endurance

“It is an important but unresolved question: Which type of training is most effective for building endurance? 1) To maintain an intensity level representing 90 percent of the maximum oxygen uptake for forty minutes. Or 2) To tax 100 percent of the oxygen uptake capacity [author’s note: Zone 4] for sixteen minutes.”

– Per Åstrand, *Textbook of Work Physiology*, 1970

A new study by Stephan Seiler may have settled this question. In it, well-trained endurance athletes used four by four minutes, four by eight minutes, and four by sixteen minutes with a two-minute rest interval training protocols. The four-by-eight-minute method showed significantly better gains in endurance than the others. Interestingly, the length of the work period turned out to select for



Philipp Reiter at his aerobic intensity while running in Yosemite National Park, California.
Photo: Jordi Saragossa

the intensity. Going as hard as one could sustain for the duration and number of repetitions resulted in the four-minute periods being done at the top of Zone 4–5. The eight-minute periods were done in Zone 4 and the sixteen-minute periods in Zone 3.

While true for the well-trained and elites level, less well-trained athletes with lower muscular endurance will not be able to sustain the high speeds/intensities for the full duration of these work periods or the full workout. For this reason we suggest the inclusion of Zone 3 intervals and Muscular Endurance training for most athletes as preparation for these Zone 4 workouts.

A LITTLE HISTORY - THE WHAT

Interval training has been around for over a hundred years in various forms, but we owe the popular, modern variation of it to one man. In the late 1930s German track coach Woldemar Gerschler coined the term *interval training* when describing his method of intermittent training where high-intensity work bouts were separated by intervals of rest. His theory was that it was during these intervals of rest that the most important changes to the heart muscle occurred that resulted in an increase in stroke volume. Today, the word *interval* is normally used to describe the work bouts rather than the rest periods. The changes to the heart's stroke volume that Gerschler saw are the





reason that Zone 4 intervals are often referred to as VO_2 max training. Recall that stroke volume is the main limiter to VO_2 max.

Gerschler and his fellow researchers proposed that by breaking the competition distance down into short segments, repeating several of these relatively short bouts of high-intensity work separated by rest intervals, the athlete could accomplish a larger volume of high-intensity training, with its cardiovascular benefits, than could be obtained from a continuous bout of exercise of similar duration. Doing so resulted in unique adaptations, and this gave birth to structured intermittent training methods.

The basic principle of interval training is that you go fast, then recover, then repeat. Gerschler stipulated a very strict procedure for the application of his interval methods, but in the intervening years many variations of how many repetitions, the duration of the repetition, the intensity of the repetition, the length of the recovery interval, and whether active or passive rest have been used. Hundreds of studies have been done on interval training to determine what works best. The ultimate answer is ... it depends. Recall Per Åstrand's quote at the beginning of this section.

It depends on:

- What training effect is desired
- What training background the athlete has
- When during the training cycle the interval training occurs

Interval training is an important component of the overall endurance training program. It will yield the best long-term results when imposed on a solid aerobic base (a high AeT) and done in conjunction with a high proportion of purely aerobic training. But most importantly, it should not be considered a shortcut to fitness, nor a stand-alone method of training.

THE HOW

"I think of my high-intensity training as falling into a few different, simple types: VO_2 max / Z4, muscular endurance / Z3, and strength like hill sprints (see below)."

– Kilian Jornet

We can make a rough division of the use of intervals into aerobic intervals and anaerobic intervals.

- Aerobic intervals are meant to maximally stress the capacity of the aerobic system (both carbohydrate and fat fueled). The effect is decided by the intensity of the work periods that are done in Zones 3 and 4.



Go hard, rest, repeat. Alyson Dimmitt Gnam catching her breath on top of a ridge in the Lewis Range, Glacier National Park, Montana. Photo: Steven Gnam

- Anaerobic intervals serve to maximally develop the anaerobic capacity and specific strength. The intensity must be very high: Zone 5. To be effective, you need to be recruiting the highest-power (FT) muscle fibers. These fibers fatigue quickly and need a long recovery. You need to be well rested and take long rest intervals between the individual work bouts. Don't do these with tired legs. They are used sparingly for the events covered in this book.

Many people mistakenly feel that these should be exhausting workouts where you drive yourself to nearly complete system failure. To be sure, this is very *hard* training and it will be tiring. But if you are reaching far beyond your current capacity for this sort of work, you may push yourself so far into fatigue that you need extra recovery time or even days off. Doing this too often can mean a loss of continuity of training.

Before we get into the details of discussing workout makeup, we're going to review the basics of interval training. We'll start with the aerobic intervals since, for most endurance athletes, these will predominate the interval volume.



Skimo training on the Passo del Tonale, Trentino, Italy. Photo: Federico Modica

AEROBIC INTERVALS

These have a powerful effect on improving aerobic endurance and can be separated into two realms: Zone 3 intervals, often called threshold intervals, and Zone 4 intervals, referred to as VO_2 max or aerobic power intervals. Both should be done using sport-specific methods.

Threshold Intervals (Zone 3)

The purpose of doing these intervals is to raise the LT speed and to increase the duration that it can be maintained. They range from twenty to sixty minutes of hard work, depending on the development stage of the athlete. These have a strong muscular endurance training effect.

If you are new to this training method, start with short-duration repetitions such as three to six minutes. Use short, active recoveries in the ratio of 3:1 or 4:1 work to rest. Experienced endurance athletes may work up to 3x20 or even 2x30 minutes for one of these sessions.

Since this is a fun sort of hard and feels like training, some athletes inadvertently include some in almost every session. As we hope you understand by now: there is not one workout or one intensity that can optimally achieve what the combined effect of several different intensity doses and durations can achieve. While threshold interval workouts are an important component of the training

repertoire, they make up a small percentage of the overall volume on a yearly basis for the most successful endurance athletes.

Aerobic Power Intervals (Zone 4)

Aerobic power (often called VO_2 max) intervals are conducted at intensities greater than LT; usually at near 95 percent of max heart rate. Total volume of the work bouts can range from fifteen to thirty minutes. These workouts maximally utilize the capacities of all the systems involved in endurance. The effectiveness of these workouts is very reliant on the athlete's muscular endurance, which is why, in the Planning Section we recommend using a period of special muscular endurance work and Zone 3 training to build sufficient muscular endurance to make these Zone 4 intervals more effective. If your leg muscles become tired and you are forced to slow down, causing the heart rate to fall below the Zone 4 range, you are no longer getting the intended benefits of Zone 4. This is why, for best effect, these workouts need to be done when you are rested. The less fit you are, the shorter the work periods should be. Muscular endurance will be discussed in more depth in the Strength Section.

When you are comfortable with Zone 3, meaning you can manage around 10 percent of the total weekly time in Zone 3, begin to add Zone 4. Do this by replacing some of the Zone 3 time with Zone 4 time. Replace two minutes of Zone 3 with one minute of Zone 4. If you struggle to recover and handle this shift, reduce the volume of Zone 4.

30/30s

These are a good way to introduce Z4 intensity into your interval routine and add in a bit of speed to what may become some fairly slow but hard uphill efforts. You'll need a good warm-up that culminates in three to four minutes at Z3. Then you'll be ready to roll right into the 30/30 sets.

Thirty seconds is not very long to run or ski hard, so you'll need to control your pace so that you don't start out too fast and then be forced to slow dramatically after the fifth one. The best way to do this is to build your effort and heart rate to where you are hitting Z4 by the middle of the second 30/30 repetition. Then slow for thirty seconds—but only enough to allow your heart rate to drop five to seven

beats. Repeat this until you feel the power and speed drop or until you notice that you can't get your heart rate into Z4. As your endurance improves, you'll be able to increase the volume of this workout.

We like to start athletes with 2x6 min of 30/30s with a couple minutes' easy between the sets so they can get the feel of the pace and effort. Next session, go to 2x8 min, then 2x10 min until you are doing 2x15 min, then go for one twenty-minute session. Thirty minutes' total time in one bout is a good goal but tough to achieve. These are especially useful for skimo, where speed and turnover need to be trained along with endurance.

When at this stage in your training progression, do the Zone 3 and Zone 4 workouts on separate days and allow at least seventy-two hours between them.

Work periods range from thirty seconds up to eight minutes with a work-to-rest ratio of 1:1 up to two minutes, two minutes is adequate for longer work periods. Rest intervals should be active and not full rest so that the aerobic system stays revved up.

MUSCULAR ENDURANCE

Muscular endurance is a special form of strength training used to increase the endurance of the faster-twitch/higher-power muscle fibers that are responsible for the limits of endurance performance. This type of training is done by imposing extraordinary loads on the muscles, which causes local muscle fatigue in the specific locomotion muscle groups rather than a global fatigue. You can do this by either using steep hills and/or by adding weight to your body, or special gym workouts discussed later. This method can be employed throughout the Base Period. Since it is a form of strength training, this method is covered in detail in the strength section on page 199.

ANAEROBIC INTERVALS

You may ask why athletes involved in the sports we cover in this book need to concern themselves with increasing what seems like a quality that only sprinters need. After all, haven't we spent the last hundred-odd pages emphasizing the importance of the aerobic system? Anaerobic capacity training in the methods described below are actually a form of very specific strength training. It turns out that improving this type of sport-specific strength also improves *economy*. Improved economy means lowered energy cost at any given speed. This can have major performance implications over multihour events. Anaerobic intervals will make up only a tiny but important fraction of any endurance athlete's overall training.

Increasing anaerobic capacity is a slow process because the amount of work that an athlete can handle in this realm is quite small compared to the daily and sometimes twice-daily aerobic workouts. This has led many coaches to observe that "Sprinters are born and distance runners are made." Translation? Almost everyone will respond positively to endurance training, whereas speed and power is largely a genetic gift. Therefore, responses to this training will vary greatly. It used to be thought that raising the anaerobic capacity was impossible, but more recent long-term observations indicate that it can be raised with judicious training over several years. We're not going to turn you into the next Usain Bolt with this training, but the work will improve your uphill speed in a way that complements the endurance training.

Training to increase anaerobic capacity uses the same basic format as an aerobic interval session: go hard, rest, repeat. But in this case you will take a longer



Kilian Jornet, clipped into a fixed line, negotiates a ridge on day four of the 2017 Pierra Menta. Kilian went on to take second place that year. France. Photo: Jocelyn Chavy

recovery between repetitions. The distinction from aerobic interval training is that these exercises are not meant to train endurance, so it is important that the anaerobic system is completely (or very nearly) fully restored so that each repetition can be executed at the same maximum intensity. This requires very short work periods of eight to fifteen seconds separated by long recovery times of at least one to two minutes if the intensity of the reps is in the correct intensity range. A rest-to-work ratio of at least 4:1 or 5:1 and up to 10:1 is used. If the rests are too short, the workout will shift to training endurance because the high-power muscles you are trying to train, which have poor endurance, will not recover and will drop out of the recruitment pool. Then the more endurance-trained motor units take over. The result will be slower reps and less anaerobic capacity/strength improvement.

Hill Sprints/Hill Bounding and Downhill Speed

A very effective anaerobic capacity, leg power workout we like to use is Hill Sprints or Hill Bounding once a week during the Base Period. These are repetitions of very short duration (eight to ten seconds) with complete recovery (one to two minutes). The training effect in the early workouts will be mainly



Fig. 5.3 Hill Bounding

Hill Sprints are exactly what the name implies: a sprint uphill with normal (albeit modified for the steeper grade) running technique.

neurological as your brain learns how to sprint. In our experience, most endurance athletes struggle with these workouts. The first several of these workouts may not even make you tired, so they don't feel like training and you'll wonder, *Why am I doing this?* The rest periods feel way too long, so endurance athletes want to shorten them. If you are not getting tired with these, it's because you are mainly using your slow-twitch motor units and they are so well endurance trained that they do not get tired in eight to ten seconds. *If it feels ineffective, then that's a sign that you need it even more.* As your brain learns to recruit more fast-twitch fibers, you will become more powerful. This will translate to longer stride lengths and improved economy.

Learning to run and ski downhill fast is like learning any skill: it takes practice. Both these skills take a lot of leg strength and agility. A good way to approach their development is to insert short (one to two minutes) high-speed downhill repetitions into some of your weekly easy-distance workouts. Increase the number and duration of these reps over the course of weeks.

Hill Bounding is done with a bounding stride where you exaggerate the length of the stride and spend more time in the air with both feet off the ground and with a lower-tempo leg turnover than the hill sprint running. This can be done with poles by skiers and without by runners.



Anton Krupicka runs a rocky ridge in the Tetons, Wyoming. Photo: Fredrik Marmsater

Hill Sprint/Bounding Guidelines:

Use a 30 percent grade hill or steep stairs. Steeper is generally better for building strength and power, but hills over 50 percent can be hard to maintain traction on. Stairs work best when you get very steep (if you can find steep and long stairs). Skip stair steps to create a bounding workout. Follow the long warm-up procedure outlined below to avoid injury. If you are inexperienced with this type of sprint training, ease into this program by starting the workouts at 80–85 percent of maximum effort and slowly increasing the output during the repetitions for the first two to three workouts. Injury is a real risk when using this program.

Special considerations: The grade of hill/stairs in percent is a ratio of the change in vertical distance divided by the change in horizontal distance. A 30 percent (about seventeen degrees) grade rises three feet or three meters for every ten feet or ten meters of horizontal run. Long sets of stairs that will take you ten to twelve seconds to sprint up can be hard to find. The typical fire escape stairs in commercial buildings are usually in the range of thirty to forty degrees, making them an ideal gradient except that they have landings spaced about ten feet (three meters) apart. Regardless, they are fine to use if nothing else is available.

HILL SPRINT/BOUNTING WORKOUT

Warm-Up

This is very important!

1. Twenty minutes of running warm-up with a gradually increasing pace finishing with two to three minutes at a Z3 effort. Break a sweat during this run. This gets your muscles warmed up and the aerobic system going.
2. Ten minutes of dynamic stretching: bouncing on toes, lunges, legs swings, and ballistic toe touches. This activates the stretch reflex in the tendons to let them know they are about to get a workout.
3. 2x20 or 30 sec medium-to-hard effort hill run or bound. These wake up the anaerobic energy system and the associated fast-twitch muscle fibers you need to do this sort of work. One to two minutes' slow walk downhill between these.
4. 2x20 sec skipping uphill. This gently introduces the dynamic and ballistic loading that you will be working next. One to two minutes of slow downhill walking between these.

Workout

Once you feel well warmed up and have no pain, you are ready to begin the main work.

1. Execute 6–8x10 sec of maximum power bounding with a long stride or running sprints. Take a two-minute easy downhill walk as recovery between sprints and a three-minute rest between sets when and if you increase the number of repetitions beyond eight.
2. Twenty minutes' easy jog cool down.

Workout Progression

1. You will progress by adding more repetitions to the workout. Do not increase the length of the repetitions beyond ten seconds.
2. If you are new to this form of training, be cautious. As mentioned above, this is a potentially injury-causing workout. Start with using 90–95 percent of maximum force. Only add force after you have handled two to three of these workouts with no problem.
3. Add no more than two repetitions per week to the total volume.

Note:

One of the biggest problems endurance athletes have doing Hill Sprint and Bounding workouts is exactly what makes this particular training good for endurance athletes: They hate resting. This training requires a radical shift in thinking. The extended rests are a vital part of the training effect. There is a good reason why sprinters may only run a total of a couple hundred meters in a workout and spend most of their time stretching and resting on the grass. It's not that they are lazy. It's that intensity is king in these workouts.

ANAEROBIC ENDURANCE

This may seem like a misnomer. How can a power output that is sustainable for only a minute, at most, be considered endurance? For most endurance athletes, this training will make up only a tiny fraction of the yearly volume. As you will read on the next page, Kilian does only two to three of these workouts. Nonetheless, it can still play a role for any event that has surges, or where maximum outputs are required. This type of training is often misunderstood and consequently misused.

This stuff's very hard. It hurts. It requires a great deal of willpower. This is maximum training at its absolute maximum. You have to drive yourself to exhaustion to get the benefits. It has to be used judiciously. Most endurance athletes have a very low capacity for this type of work. It may take only two to three of these workouts spaced seven to ten days apart to reach the highest levels of anaerobic endurance needed for the events covered in this book. Fast-twitch athletes or born sprinters can do much more and will see bigger gains. But that is a tiny fraction of mountain athletes. This is definitely *not* a type of training that should be included in all phases of training. Less well-trained individuals should use it with caution only in preparation for important short races like the VK. The benefits are small and the risks are high.

Two of the popular interval methods for this type of training are:

- Maximal work bouts of two minutes with a work-to-rest ratio of about 2:1 repeated four to six times or until exhaustion. These will build very high levels of lactate, and the theory goes that many beneficial aerobic and anaerobic adaptations occur from this sort of workout. Due to the short recovery periods, fatigue may prematurely halt this workout, limiting the volume of training and hence its effectiveness.
- A second type involves going the maximum speed for twenty to thirty seconds with a recovery of two to three minutes, repeated eight to twelve times. These intervals allow a higher speed and with their longer recovery, may allow a greater volume to be completed in a session. As you read, this is Kilian's preferred method.



NOTES FROM KÍLIAN

Before every race season, I do two to three shorter-interval sessions for my maximum speed. These intervals are short and I go as fast as I can on a slight hill. These are bouts of twenty, for a maximum thirty seconds. I will do ten rounds, then rest for a bit, then do another ten rounds. Maybe a

third round if I still feel a lot of bounce in my legs and heart. This usually happens only in the second or third session. This helps me have that extra gear for a really short, hard effort in a race (like in a good fight with a strong competitor).

Unlike the aerobic interval methods explained earlier, the point of anaerobic intervals is to produce as much lactate as possible and to maintain that high level for as long as possible. The workouts are exhausting and require significant recovery time along with a healthy dose of aerobic regeneration work afterward.

In the Planning Section we will demonstrate how to incorporate these workouts into your own training plan.

ATHLETE STORIES

Mike Foote Sets New World Record for Uphill Vertical Skied in Twenty-Four Hours

Laura Larson

Mike Foote cranked up his final lap—his sixtieth—with fifteen friends in tow. It was nearing 9 a.m. on March 18, almost a full day after he first started skinning up and skiing down the 1,020-foot Ed's Run at Whitefish Mountain Resort in Montana. During the previous lap, he'd eclipsed the world record for vertical ascended and descended in twenty-four hours on ski-mountain-eering gear. This last push was a celebration.

"It was really emotional and really special, and I was surrounded by people that I love," he says. "It was an amazing thing." It was also his second-fastest lap of a very long day.

Foote reached the bottom with seven minutes to spare, having gained a mind-bending 61,200 vertical feet since 9 a.m. the previous day. "I could have, in theory, climbed for five more minutes and then descended again, but I chose [not to]. It felt like a very clean thing. I did a final full lap, and I was really happy being done."

The Missoula-based ultrarunner and skimo racer had been training toward this esoteric goal for several months. Austrian Ekkehard Dörschlag had set the bar at 60,000 feet on the dot in 2009, during a twenty-four-hour event at a ski resort in Austria, and once Foote stumbled across his achievement, the desire to surpass it took hold. He had been on the hunt for the skimo equivalent of a mountainous 100-miler, and here it was: an effort perfectly suited to a guy who shines in steep, technical, long-duration trail races like the Hardrock 100 and UTMB. He was going to chase some vert.

Foote began collaborating with Uphill Athlete Master Coach Scott Johnston in December 2017, and together, the two came up with a program that would have Foote—already a high-level athlete—doing the bulk

of his workouts at event pace. They calculated this to be around 2,800 to 3,000 feet per hour, which would give him the necessary buffer for transitions and descending.

"By defining the pace that he needed to maintain, that gave us a target to use. That formed the basis for the whole training strategy. It really wasn't that complicated," explains Johnston. "Mike has an incredibly high work capacity, so he managed to put in some phenomenal weeks of vertical. Some of the numbers are staggering."

Over the three months leading up to mid-March, Foote averaged 29,233 vertical feet per week and racked up a staggering 42,360 feet, 50,526 feet, and 50,942 feet during his three biggest weeks. Most weeks saw him training over twenty hours, with a peak of twenty-four hours.

"My workouts were less about intensity and more about long, slow efficiency—really holding that zone and that effort for very long periods of time and seeing how that felt," says Foote. "Skiing allows me to do higher volume in [terms of] time, because there's no impact, so my body can handle it better."

Before the record attempt, the two biggest days Foote had ever had on skis—both around 20,000 feet—were back-to-back, just a few weeks out from the event. The block went amazingly well. "I felt like I could have done it then," he says. But then a switch flipped, and his body rebelled. "I cratered afterward, and the next week I felt horrible. I had to travel that weekend, and I missed a lot of sleep and didn't really recover. Then I was in the hole."

Mike Foote halfway through the 600-mile Crown Traverse from his doorstep in Missoula, Montana, to Banff, Alberta.
Photo: Steven Gnam



That hole cast a pall over Foote's initial laps on March 17, leaving him wondering if he had overdone it in training. "I was feeling pretty junky for the first two to three hours," he admits. "I went into this very negative spot of feeling like I might let everybody down and I wasn't going to be able to do this thing I'd put so much time into and made so many sacrifices for. I was mentally in a tough spot."

A lot of it turned out to be no more than jangly nerves. Once his body eased into the activity, he fell into a comfortable rhythm. "I was just trying to stay super-consistent," Foote says. "I wasn't really looking at my watch or anything, I was just trying to ski by feel."

That methodical approach carried over to the descents, which he used for recovery while still prioritizing speed. "It was important for me to not take the downhill for granted. I definitely pushed every single descent to find that balance between going down really, really fast but at the same time not trashing myself," he says.

Ever mindful of time, he streamlined the transition process by setting his base-area crew up with an additional pair of Dynafit skis and several pairs of skins. When he'd get to the bottom, he'd click out of one pair of skis and click right into another pair, prerigged with skins.

As he churned out lap after lap, Foote fueled steadily, consuming everything from gels and Coke to real food like soup broth, PB&Js, sweet potato muffins, and sticky rice balls with bacon. "I was able to eat about 400 calories an hour, which is just a ton," he says. "I also went through two-thirds of a quart of maple syrup, which I'm really proud of."

By the twelve-hour mark, he'd already racked up 33,500 feet. Plus some fans. The North Face tent stationed in the middle of the run during the day proved to be a draw, attracting numerous skiers curious to see what was going on. (Foote is a TNF Athlete; the company fashioned a custom skimo suit for him for this effort.) At the bottom, these skiers became spectators and cheerleaders, giving Foote a much-needed energy boost and creating a fun, supportive atmosphere.

The weather was perfect during the day: forty degrees and sunny. But when the temperature dropped into the twenties that night, the sloppy daytime mashed potatoes froze up, rendering the surface conditions extremely tricky. "I was doing a lot of sliding backward on my skins," Foote recalls. "I had to really yard on my poles and straps just to push myself

up the hill." (A few days out, he still has numbness in his left hand from that overreliance on his straps.)

To make matters even more challenging, he battled a downslope wind for six-plus hours that night, and his quads and feet were killing him on the icy, chattery skis down.

"I was literally yelling out loud on the descents when I would hit a chunky section. I've never experienced anything like that," he says. His feet were battered from being stuck in the same sweaty ski boots for the full twenty-four hours—a decision he regrets. "When I finally took my boots off, it was not a pretty sight."

During the night, his 3,000-foot lead on the record eroded to 1,000 feet—a mere twenty minutes. He knew he needed to dig deep. "I was thankful for the pacers, because they were keeping me honest, but I was still going pretty slow." Among those pacers was fellow ultrarunner Luke Nelson, who stuck with Foote for six hours. "Luke cracked the whip on me in a big way and was an awesome, awesome support out there," he says.

By morning, Foote had maintained his gap on the record, and he finished strong, surrounded by that gaggle of pals. He was reawakened to the power of simply moving forward, of putting one foot in front of the other. "You'll come out the other end. I always forget that, and then I'm reminded during something like a 100-mile mountain race or this effort," he says. "It's a really unique experience to feel so close to giving up, so close to quitting, and then to be able to focus on the movement—on relentless forward progress, and then suddenly your body and your mind come back around."

Absolutely integral to his success was his crew of twenty-odd friends, including his girlfriend, as well as a string of other unexpected helpers along the way. These included people like the groomers who, unbidden, started coming by to put down fresh paths of soft corduroy for him during the night, making those teeth-chattering descents a little less brutal.

"No way could I have done this alone," Foote insists. "It's a reminder that to reach your best moments, it's important to have support." Along with pacers and folks wrangling gear at the transitions, he had people walking beside him handing him food and others taking care of all the documentation necessary for a world-record attempt—photos, lap times, everything to ensure the twenty-four hours would hold up under scrutiny.

"It was a full-on team effort." ■

Laura Larson is a writer and editor whose career has taken her from outdoor publishing to celebrity gossip and back to outdoor publishing. She splits her time between proofreading how-to and where-to guides for

Mountaineers Books, writing for *Uphill Athlete*, and trying to keep up with her dog on the local trails. He deserves a lot of the credit for her two 50-mile and six 50K finishes since 2015.



Mike Foote in the daylight hours of his record-setting attempt for human-powered vertical ascent and descent on skis: 61,170 feet in 24 hours. Whitefish, Montana. Photo: Steven Gnam

